Individual Weekly Report – Week 3

Data Cleaning

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Department	Artificial Intelligence and Data Science
Project Title	Youth Smoking and Drug Consumption: From Raw Data to Visual Insights
Role	Data Visualization and Dashboard Development
Week	Week 3 – Data Cleaning

Objective of the Week

The aim of this week was to support the data cleaning process. Since raw data often has problems, I used visualizations to see these problems clearly instead of only reading numbers in a table.

Work Done

Creating initial visualizations: I created basic graphs for the main variables to check the overall quality of the data. These first graphs gave a quick view of how the data looked before any cleaning was done.

Checking distribution and unusual values: I used graphs to understand how the values were spread. Values that were far away from the rest or that looked unusual were easy to notice in the graphs, and I marked them for the team to review.

Observing missing values and inconsistencies: Through visual analysis, I observed where values were missing and where the data was not consistent. Showing this visually made it easier to explain the problems to the team.

Discussing improvements with the team: I discussed my observations with team members and talked about how the visualizations could be improved to show data quality issues more clearly. These discussions also helped the team decide how to handle the problem areas.

Tools and Techniques Used

Basic graphs for distributions, visual checks for missing and unusual values, and team discussions.

Outcome of the Week

The visualizations pointed out the data quality issues that needed attention. This helped the team clean the data with a clearer understanding of what had to be fixed.

What I Learned

I learned that visualization is not only for the final result. It is also a useful tool at the data cleaning stage, because problems that are hard to see in a table are easy to see in a graph.

Contribution to the Project

My graphs supported the team's data cleaning work by showing the issues in the data in a simple visual form.

Challenges Faced

It was sometimes hard to decide whether an unusual value was a real feature of the data or a mistake. I handled this by marking such values in the graphs and discussing them with the team before any decision was taken.

Plan for Next Week

In Week 4, I will create histograms, boxplots and scatterplots to study the data deeply and understand its patterns and relationships.

Weekly Summary

In Week 3, visualization worked as a quality check for the data. The graphs made problems like unusual values and missing data easy to see and discuss.